Forward Deployed Engineer (FDE) Role Map

Core Identity & Mission

A Forward Deployed Engineer (FDE) operates at the intersection of full-stack systems engineering, customer delivery, and product strategy. Embedded directly with strategic enterprise accounts, FDEs translate ambiguous business needs into scalable, production-grade software and AI agent architectures while feeding field patterns back to internal product and research teams.

Core Responsibilities, Skills, & Customer Scenarios

Operational Pillar	Primary Responsibilities	Technical & Soft Skills Required	Typical Customer Scenarios
1. Architecture & AI Systems Engineering	• Design end-to-end agent architectures (LLMs, RAG, memory, tool integration). • Build MCP servers, sub-agents, and execution services. • Integrate AI tools into mission-critical workflows.	• **Languages:** Python, TypeScript, SQL, Java/Scala. • **AI Tech:** LLM APIs, vector DBs, orchestration, prompt engineering. • **Systems:** Distributed systems, microservices, system design.	A healthcare client needs an AI agent to parse complex medical records and interface safely with legacy EHR systems while maintaining low latency and high accuracy.
2. Production Delivery & Infrastructure	• Develop full-stack web applications and data pipelines. • Deploy on enterprise cloud infrastructure (AWS/Azure/GCP). • Maintain security, access controls,	• Cloud-native infra (Docker, Kubernetes, IaC, CI/CD). • Enterprise data processing & pipeline maintenance. • System	A financial services firm requires a customized data integration pipeline and full-stack dashboard to monitor fraudulent transactions across terabyte-scale datasets.

	and compliance.	observability, logging, and security frameworks.	
3. Quality, Evals & Observability	• Implement LLM quality harnesses (LLM-as-a-Judge, golden datasets). • Track tracing, guardrails, latency, and retries. • Establish human-in-the-loop oversight workflows.	• Offline benchmarks & online experimentation. • Observability, regression testing, and telemetry tracking. • Fallback and graceful degradation strategies.	An enterprise deployment suffers from model hallucinations and quality regressions following an upstream LLM release; the FDE must deploy eval harnesses and guardrails to restore reliability.
4. Customer Immersion & Technical Leadership	• Embed directly with client engineering and executive teams. • Scope ambiguous requirements into structured backlogs. • Conduct architecture reviews and technical workshops.	• High agency & navigating organizational ambiguity. • Strong written & verbal technical communication. • Dual fluency: bridging technical ICs and executive leadership.	A government defense partner asks an open-ended operational question ("How do we optimize supply chains in real time?"); the FDE scopes the technical architecture and builds the initial production solution.
5. Product Feedback & Asset Reusability	• Codify customer solutions into reusable frameworks/accelerators. • Share operational	• Pattern recognition across diverse domain deployments. • Internal cross-functional	Multiple enterprise accounts request similar custom tool-calling functionality; the FDE abstracts the deployment code

	telemetry and field feedback with Product & Research.	collaboration (SWE, Product, Research).	into a core platform asset to accelerate future client rollouts.